

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination May 2017

PH117/PH125 Human Anatomy and Physiology II

Date: 12.05.2017, Friday

Time: 10:00a.m. to 1:00p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Which part of brain is having an important link with pituitary gland?
[A] Hypothalamus
[B] Thalamus
[C] Cerebrum
[D] Cerebellum
 2. Which organ of digestive system is also an endocrine gland?
[A] Liver
[B] Small intestine
[C] Large intestine
[D] Pancreas
 3. Which of the following hormone act on nuclear receptor?
[A] Insulin
[B] Corticosteroid
[C] Thyroxine
[D] Adrenaline
 4. Which of the following hormone is involved in process of lactation?
[A] Growth hormone
[B] Thyroid stimulating hormone
[C] Prolactin
[D] Follicle stimulating hormone
 5. Acromegaly is caused by excess levels of _____
[A] Growth hormone
[B] Thyroid stimulating hormone
[C] Prolactin
[D] Follicle stimulating hormone
 6. Depolarization phase of neuronal action potential is occurred due to rapid influx of _____.
[A] Na⁺
[B] K⁺
[C] Ca²⁺
[D] Cl⁻

7. Which part of the brain consist of vasomotor center and involved in maintenance of blood pressure?
 [A] Cerebrum
 [B] Medulla Oblongata
 [C] Thalamus
 [D] Epithalamus
8. Which of the following autacoid is predominantly involved in the allergic reaction?
 [A] 5- Hydroxytryptamine
 [B] Histamine
 [C] Prostaglandins
 [D] Dopamine
9. Which type of cells are involved in bone building process?
 [A] Osteogenic cells
 [B] Osteoblasts
 [C] Osteocytes
 [D] Osteoclasts
10. _____ reduces friction and absorbs shock at freely movable joints.
 [A] Diaphysis
 [B] Metaphysis
 [C] Metaphysis
 [D] Articular cartilage
11. Which of the following hormone inhibits the activity of osteoclast, speeds blood Ca^{2+} uptake by bone, and accelerates, Ca^{2+} deposition into bones?
 [A] Parathyroid hormone
 [B] Calcitonin
 [C] Calcitriol
 [D] Corticosteroids
12. Synovial fluid functions to (1) absorb shocks at joints, (2) lubricate joints, (3) form a blood clot in a joint injury, (4) supply oxygen and nutrients to chondrocytes, (5) provide phagocytes to remove debris from joints.
 [A] 1, 2, 4, and 5
 [B] 1, 2, 3, 4, and 5
 [C] 1, 2, and 3
 [D] 3 and 4
13. Which of the following statements is *false*?
 [A] The first neuron in an autonomic pathway is the preganglionic neuron.
 [B] The postganglionic neuron's cell body is within the CNS.
 [C] Postganglionic neurons relay impulses from autonomic ganglia to visceral effectors
 [D] All somatic motor neurons release acetylcholine
14. How many pairs of cranial nerves are linked with Central Nervous System?
 [A] 11
 [B] 12
 [C] 13
 [D] 14
15. Which of the following cranial nerve constitute 80% of parasympathetic motor outflow?
 [A] Cranial Nerve III; Oculomotor Nerve
 [B] Cranial Nerve VII; Facial Nerve
 [C] Cranial Nerve IX; Glossopharyngeal nerve
 [D] Cranial Nerve X; Vagus Nerve

16. Which of the glial cells are involved in the process of myelination?
 [A] Astrocytes
 [B] Oligodendrocytes
 [C] Microglia
 [D] Satellite cells
17. Which one of the following is the primary source of progesterone in the later stages of pregnancy?
 [A] Foetus
 [B] Endometrium
 [C] Corpus Luteum
 [D] Placenta
18. Which of the following is not paired in the male?
 [A] Epididymis
 [B] Ejaculatory duct
 [C] Urethra
 [D] Vas deferens
19. Which hormone of hypothalamus controls ovarian and uterine cycles?
 [A] GHRH
 [B] TSH
 [C] GnRH
 [D] FSH and LH
20. The female structure that is homologous to the testis is the
 [A] Ovary
 [B] Uterus
 [C] Vagina
 [D] Clitoris

SECTION – II

- Q 2** Attempt any **TWO** of the following
- | | | |
|----------|--|-----------|
| A | Classify joints. Write a note on synovial joints. | 05 |
| B | Classify hormones according to their chemical nature. Enumerate endocrine gland alongwith hormones secreted by them. | 05 |
| C | Write a note on Cranial nerves. | 05 |
- Q 3** Attempt any **TWO** of the following
- | | | |
|----------|--|-----------|
| A | Differentiate: Autonomic nervous system and Somatic Nervous system | 05 |
| B | Give comparison between nervous system and endocrine system. | 05 |
| C | Compare the major events of the ovarian and uterine cycles | 05 |

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- | | | |
|----------|--|-----------|
| A | Draw neat labeled Diagram of Neuron. Explain physiology of Nerve action potential. | 05 |
| B | Write the physiological role of Histamine and Prostaglandin. | 05 |
| C | Discuss in detail about Pancreas, its hormones and functions. | 05 |
| D | Write about the structure and functions of skeleton. | 05 |

- E** Draw a neat and labelled diagram of Male reproductive system and Discuss Spermatogenesis in brief. **05**
- F** Write a note on CSF. **05**
- Q 5** Attempt any **FOUR** of the following
- A** Describe the functions of Hypothalamus and medulla oblongata. **05**
- B** Explain physiology of anterior pituitary gland . **05**
- C** Explain functions of sex hormones. **05**
- D** Explain the following terms:
1. Diabetes mellitus
 2. Rheumatoid Arthritis **05**
 3. Hyperthyroidism
 4. Parkinson's Disease
- E** Enumerate different parts of brain and write the functions of Cerebrum. **05**
- F** Discuss the role of calcitonin and parathyroid hormone in the maintenance of blood Calcium levels. **05**
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